



Maintenance Schedule

Diesel engine

V4000R41x

V4000R42

Application group 2A

M050640/08E



A Rolls-Royce
solution

Engine model	kW/cyl.	Application group
8V4000R41R	106 kW/cyl.	2A; Continuous operation, unrestricted
8V4000R41	125 kW/cyl.	2A; Continuous operation, unrestricted
8V4000R41L	138 kW/cyl.	2A; Continuous operation, unrestricted
12V4000R41R	115 kW/cyl.	2A; Continuous operation, unrestricted
12V4000R41	125 kW/cyl.	2A; Continuous operation, unrestricted
12V4000R41L	138 kW/cyl.	2A; Continuous operation, unrestricted
16V4000R41R	106 kW/cyl.	2A; Continuous operation, unrestricted
16V4000R41	125 kW/cyl.	2A; Continuous operation, unrestricted
16V4000R41L	138 kW/cyl.	2A; Continuous operation, unrestricted
20V4000R42	135 kW/cyl.	2A; Continuous operation, unrestricted

Table 1: Valid for

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer MTU.

Definition of MTU

MTU refers to Rolls-Royce Power Systems AG and MTU Friedrichshafen GmbH or an affiliated company pursuant to Section § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture).

Maintenance concept

Information about the latest maintenance schedule is available at: <http://www.mtu-solutions.com>.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of MTU. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (hrs)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the engine is to remain out of service for a longer time period, MTU recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

1.2 Allocation matrix application group 2A

Note:

TBO identifiable via cylinder power (shown on engine ID plate or in 'Engine Data' in Business Portal, power/ number of cylinder) and, where appropriate, in contractually agreed load profile.

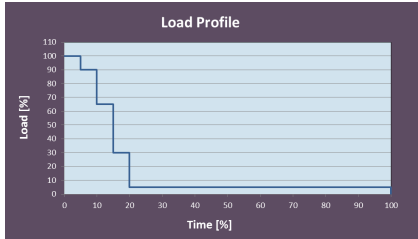
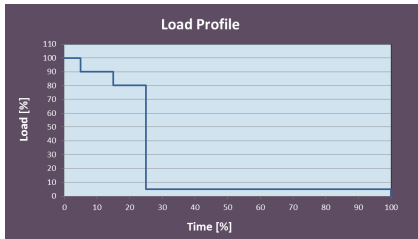
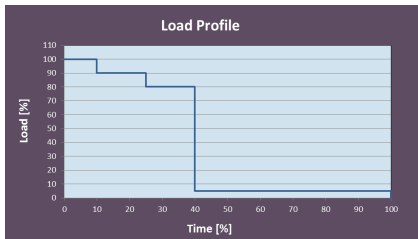
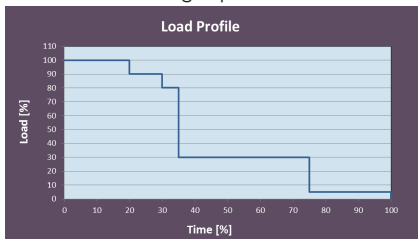
Load profile		4000 R41x; R42						
Load %	Time %	Load factor %	Load indicator %	From 0 to 106 kW/cyl.	From 107 to 125 kW/cyl.	From 126 to 138 kW/cyl.		
110 ₍₁₁₀₋₁₀₀₎	—	From 0 to 18	From 0 to 7	42000	36000	36000	—	Switcher 
100 ₍₁₀₀₋₉₀₎	5							
90 ₍₉₀₋₈₀₎	5							
80 ₍₈₀₋₆₅₎	—							
65 ₍₆₅₋₅₀₎	5							
50 ₍₅₀₋₃₀₎	—							
30 ₍₃₀₋₀₎	5							
5 _(idle)	80							
110 ₍₁₁₀₋₁₀₀₎	—	From 19 to 26	From 8 to 10	42000	36000	30000	—	Multi Purpose 
100 ₍₁₀₀₋₉₀₎	5							
90 ₍₉₀₋₈₀₎	10							
80 ₍₈₀₋₆₅₎	10							
65 ₍₆₅₋₅₀₎	—							
50 ₍₅₀₋₃₀₎	—							
30 ₍₃₀₋₀₎	—							
5 _(idle)	75							
110 ₍₁₁₀₋₁₀₀₎	—	From 27 to 38	From 11 to 17	36000	30000	27000	—	Standard 
100 ₍₁₀₀₋₉₀₎	10							
90 ₍₉₀₋₈₀₎	15							
80 ₍₈₀₋₆₅₎	15							
65 ₍₆₅₋₅₀₎	—							
50 ₍₅₀₋₃₀₎	—							
30 ₍₃₀₋₀₎	—							
5 _(idle)	60							
110 ₍₁₁₀₋₁₀₀₎	—	From 39 to 46	From 18 to 25	30000	27000	24000	—	High Speed 
100 ₍₁₀₀₋₉₀₎	20							
90 ₍₉₀₋₈₀₎	10							
80 ₍₈₀₋₆₅₎	5							
65 ₍₆₅₋₅₀₎	—							
50 ₍₅₀₋₃₀₎	—							
30 ₍₃₀₋₀₎	40							
5 _(idle)	25							

Table 2: Allocation matrix application group 2A

1.3 Noncyclical maintenance tasks

No.	Qualification	Task	Maintenance tasks	Reference task	Offset	Value	Unit	Option
0001	QL1	WM00094	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.		0	1000	HRS	
Reference task: Name of the maintenance task the offset refers to Offset: Interval of a maintenance task which depends on another interval Value: Fixed value for binding execution of a maintenance task								

1.4 Maintenance Schedule Matrix 42000 HRS

500 to 15000 Operating hours

Task	Limit	Operating hours [HRS]																													
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500	15000
WM00285	1 DAY																														
WM00286	1 DAY																														
WM00287	1 DAY																														
WM00288	1 DAY																														
WM00289	1 DAY																														
WM00290	1 DAY																														
WM00291	1 DAY																														
WM00881	1 MON																														
WM00067	2 YR																														
WM00116	2 YR																														
WM00126	5 YR																														
WM00128	5 YR																														
WM00063	2 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00066	9 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00149	1 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00084	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00086	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00170	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00050	1 YR						X						X						X					X							X
WM00136	1 YR						X						X						X					X							X
WM00076	3 YR						X						X						X					X							X
WM00664	3 YR						X						X						X					X							X
WM00692	5 YR						X						X						X					X							X
WM00052	18 YR						X						X						X					X							X
WM00094	18 YR		X				X						X						X					X							X
WM00961	18 YR						X						X						X					X							X
WM00962	18 YR						X						X						X					X							X
WM00119	3 YR												X											X							
WM00110	9 YR																						X								
WM00090	18 YR																						X								
WM00098	18 YR																						X								
WM00283	18 YR																												X		
WM00976	18 YR																												X		
WM00008	6 YR																														
WM00065	9 YR																														
WM00054	18 YR																														
WM00056	18 YR																														
WM00062	18 YR																														
WM00093	18 YR																														
WM00096	18 YR																														
WM00921	18 YR																														

DAY = Days

MON = Months

YR = Years

HRS = Hours

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Task	Limit	Operating hours [HRS]																														
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500	15000	
WM00970	18 YR																															
WM00971	18 YR																															
WM00973	18 YR																															
WM00974	18 YR																															
WM00981	18 YR																															
WM00006	6 YR																															
WM00011	18 YR																															
WM00072	18 YR																															
WM00087	18 YR																															
WM00091	18 YR																															
WM00980	18 YR																															
WM00009	15 YR																															
WM00001	18 YR																															
WM00002	18 YR																															
WM00045	18 YR																															
WM00061	18 YR																															
WM00070	18 YR																															
WM00088	18 YR																															
WM00092	18 YR																															
WM00095	18 YR																															
WM00099	18 YR																															
WM00100	18 YR																															
WM00103	18 YR																															
WM00104	18 YR																															
WM00108	18 YR																															
WM00111	18 YR																															
WM00127	18 YR																															
WM00137	18 YR																															
WM00139	18 YR																															
WM00353	18 YR																															
WM00439	18 YR																															
WM00925	18 YR																															
WM00936	18 YR																															
WM01030	18 YR																															

DAY = Days
MON = Months
YR = Years
HRS = Hours

15500 to 30000 Operating hours

Task	Limit	Operating hours [HRS]																													
		15500	16000	16500	17000	17500	18000	18500	19000	19500	20000	20500	21000	21500	22000	22500	23000	23500	24000	24500	25000	25500	26000	26500	27000	27500	28000	28500	29000	29500	30000
WM00285	1 DAY																														
WM00286	1 DAY																														

DAY = Days
MON = Months
YR = Years
HRS = Hours

Task	Limit	Operating hours [HRS]																													
		15500	16000	16500	17000	17500	18000	18500	19000	19500	20000	20500	21000	21500	22000	22500	23000	23500	24000	24500	25000	25500	26000	26500	27000	27500	28000	28500	29000	29500	30000
WM00287	1 DAY																														
WM00288	1 DAY																														
WM00289	1 DAY																														
WM00290	1 DAY																														
WM00291	1 DAY																														
WM00881	1 MON																														
WM00067	2 YR																														
WM00116	2 YR																														
WM00126	5 YR																														
WM00128	5 YR																														
WM00063	2 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00066	9 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00149	1 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00084	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00086	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00170	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00050	1 YR						X					X					X				X				X					X	
WM00136	1 YR						X					X					X				X				X					X	
WM00076	3 YR						X					X					X				X				X					X	
WM00664	3 YR						X					X					X				X				X					X	
WM00692	5 YR						X					X					X				X				X					X	
WM00052	18 YR						X					X					X				X				X					X	
WM00094	18 YR						X					X					X				X				X					X	
WM00961	18 YR						X					X					X				X				X					X	
WM00962	18 YR						X					X					X				X				X					X	
WM00119	3 YR						X										X													X	
WM00110	9 YR											X																			
WM00090	18 YR											X																			
WM00098	18 YR											X																			
WM00283	18 YR																										X				
WM00976	18 YR																										X				
WM00008	6 YR											X																			
WM00065	9 YR											X																			
WM00054	18 YR											X																			
WM00056	18 YR											X																			
WM00062	18 YR											X																			
WM00093	18 YR											X																			
WM00096	18 YR											X																			
WM00921	18 YR											X																			
WM00970	18 YR											X																			
WM00971	18 YR											X																			
WM00973	18 YR											X																			
WM00974	18 YR											X																			
WM00981	18 YR											X																			
WM00006	6 YR																														

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